

Run 224489: Wiring Fix Confirmed, First Plastic-MIP Calibration from LIQ Triples, and HV Scan 2

n_TOF EAR2 X17 — first run with the liquid scintillators (July 17, 2026)

Dylan Neff

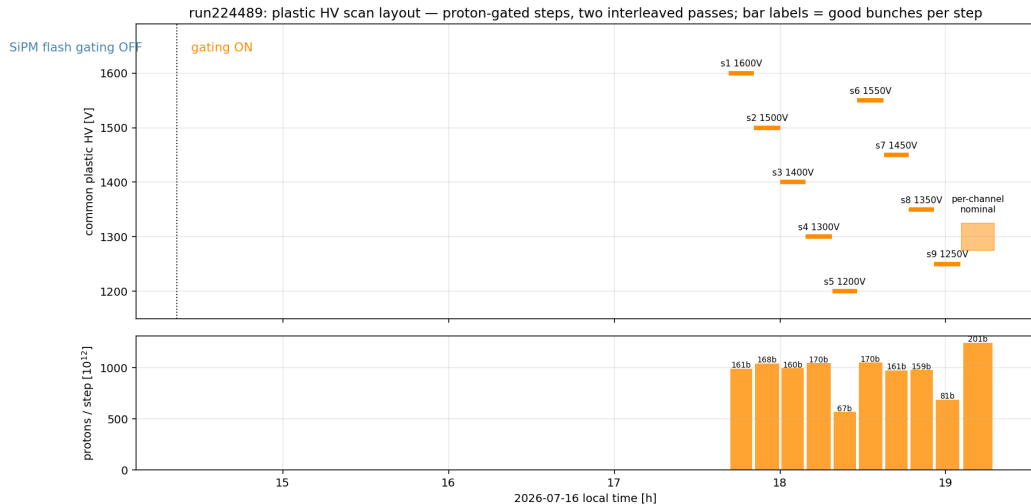
analysis and slides prepared with Claude (Anthropic AI assistant)

July 18, 2026

Run 224489: what changed since the last scan (224466)

- ▶ **Cabling fix** on the crossed A/D channels.
- ▶ **FIFO**: plastics on a linear fan-in/fan-out instead of the BNC-T split (~ 16 ns extra delay, naive $2\times$ amplitude).
- ▶ **Liquid scintillators** (LIQA-D, 1 ch/arm) in the readout for the first time: 5.3 M (C) – 21.6 M (B) hits.
- ▶ SiPM flash gating **ON all run** (no boundary).
- ▶ Same 9-step plastic HV ladder, 1200–1600 V in 50 V merged grid, + post-scan nominal-HV window.
- ▶ 1750 bunches (817 dedicated), new PSA with LIQ pulse-shape fitting.
- ▶ QA clean: **no wall outage** (0/1750; 224404 lost 46%), γ -flash $11.62\ \mu\text{s}$ on all arms, all 44 channels alive.

Scan timeline: bunches → HV steps via the CAEN log



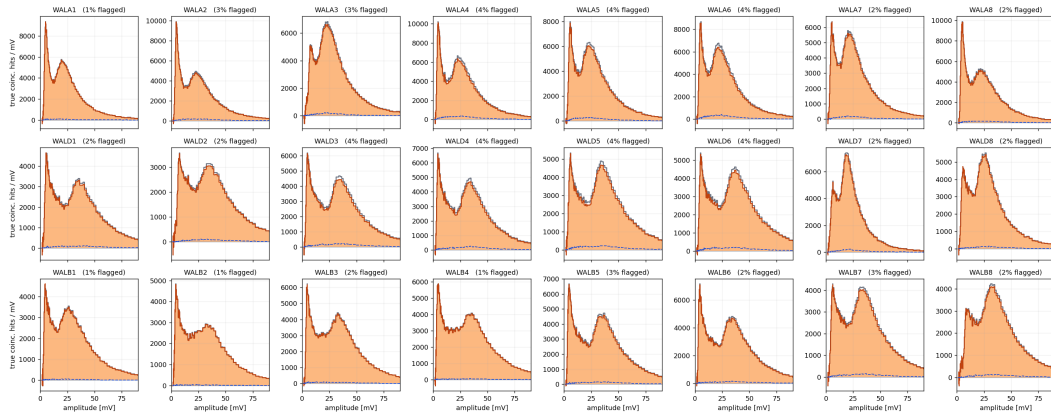
Result 1 — the cross-wiring fix is **confirmed**; A/D closed

- ▶ Duplication band (± 4 ns equal-amplitude partner, one group over): flagged fractions were **23–31%** on the crossed A/D channels in 224460 $\rightarrow$ now $\leq 4\%$ everywhere = B-arm control level.
- ▶ Wall-plastic excess now **equal on all four arms** (1.71–1.81 M within ± 10 ns) — was 6–8 $\times$ deficient on A/D.
- ▶ A/D coincident wall spectra show clean MIP bumps with **no veto**.

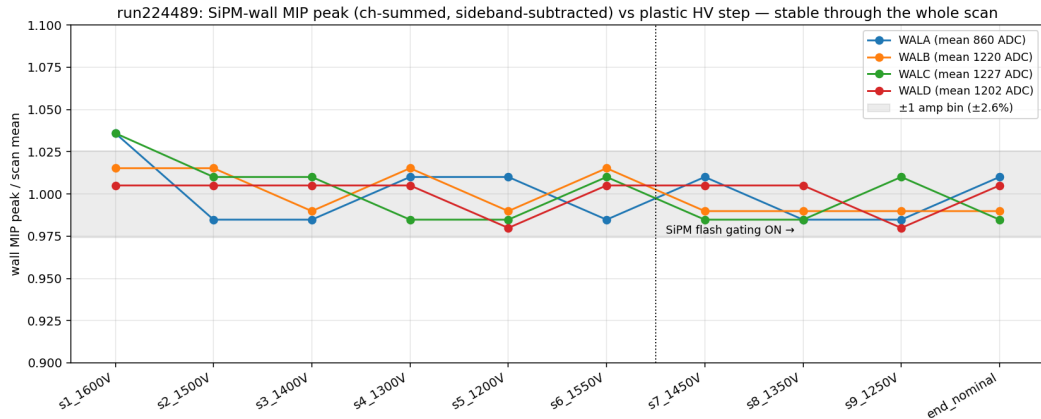
The standing A/D coincidence deficit was the cabling. Closed.

A/D duplication autopsy: before-and-after

run224489: sideband-subtracted coincident wall spectra — duplication veto (same-side neighbor within ± 4 ns, amp ratio 1/3-3); % = flagged share of signal as-is, after duplication veto, control removed (flagged) component



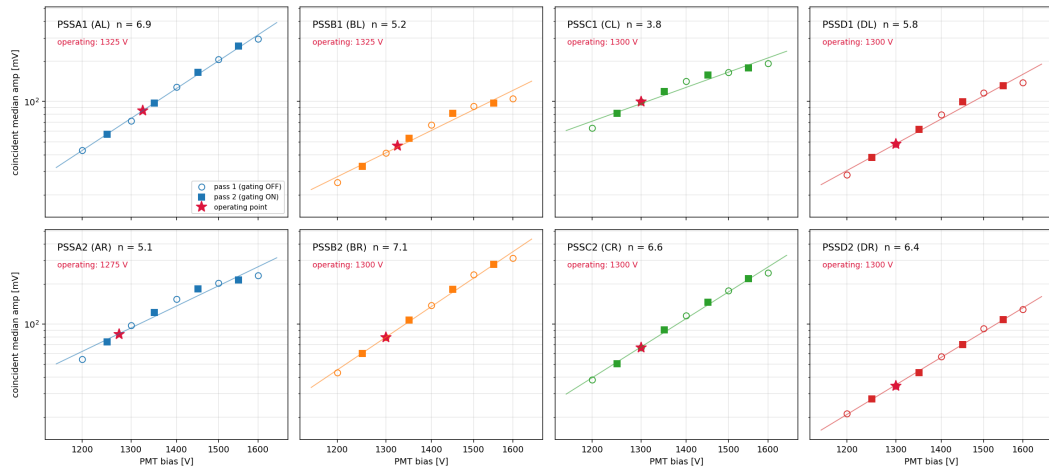
HV scan: wall MIP again immune to plastic HV (now incl. arm A)



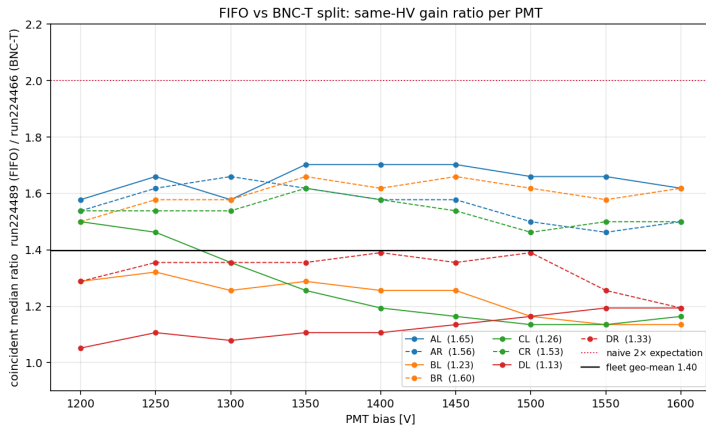
±1 bin (±2.6%) over a 3× tag-rate change ⇒ wall energy scale independent of plastic operating point.

Per-PMT gain power laws $G \propto V^n$, $n = 3.8\text{--}7.1$

run224489: plastic gain curves — wall-tagged coincident median (sideband-subtracted) vs measured HV; line = combined-pass power-law fit



Result 2 — the FIFO gain ratio is **1.4**, not 2



Same-HV coincident medians, 224489/224466: fleet geo-mean $\times 1.40$, per-PMT 1.13 (DL) – 1.65 (AL). Use the measured per-PMT ratios, not a global 2.

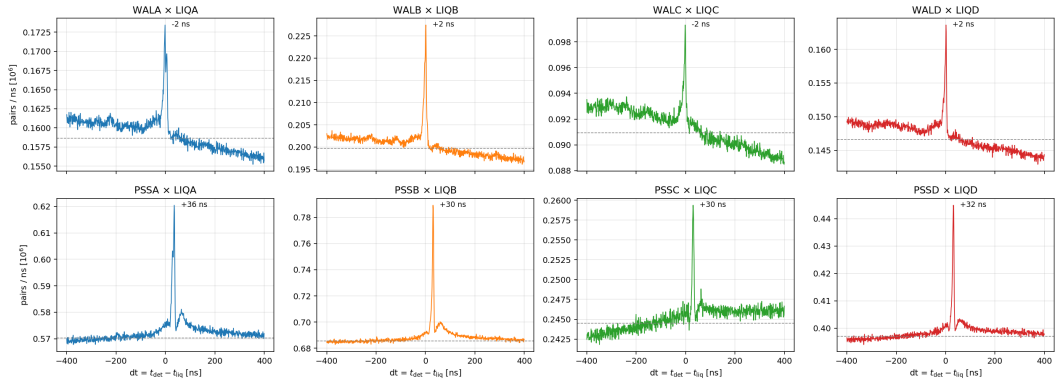
New HV equalization (supersedes 224466 table)

PMT	med [mV]	V_{sugg}	ΔV
AL	85.4	1271	-54
AR	83.6	1211	-64
BL	46.7	1409	+84
BR	79.2	1262	-38
CL	99.8	1158	-142
CR	66.5	1293	-7
DL	47.8	1368	+68
DR	34.4	1433	+133

- ▶ Target: fleet geo-mean coincident median (64.2 mV); spread now $2.9\times$.
- ▶ At equalized gains: common threshold with 99% tagged efficiency on every PMT down to **4.9 mV** (95%: 7.1 mV).
- ▶ Exported to DAQ repo (calibrations/pss/, pss_trigger/).

Result 3 — the liquids are timed in (first time)

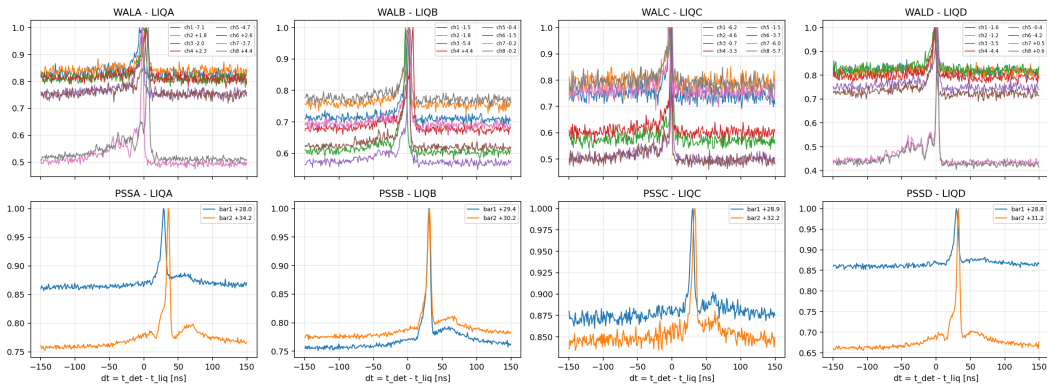
run224489: first LIQ coincidence scan (± 400 ns window) — every arm shows a sharp peak on the combinatorial floor



LIQ leads the plastic by 29.5–32.8 ns, within ± 3.3 ns of the wall; per-channel offsets (rms 3.2–4.5 ns)
→ `calib/liq_offsets_run224489.json`.

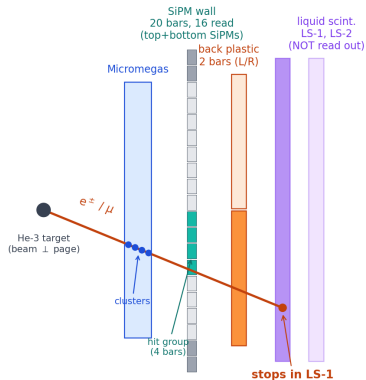
LIQ timing per wall channel / plastic bar

LIQ timing: dt peaks per wall channel (top) and plastic bar (bottom), late hits



The triple telescope: why LIQ unlocks the *plastic* MIP

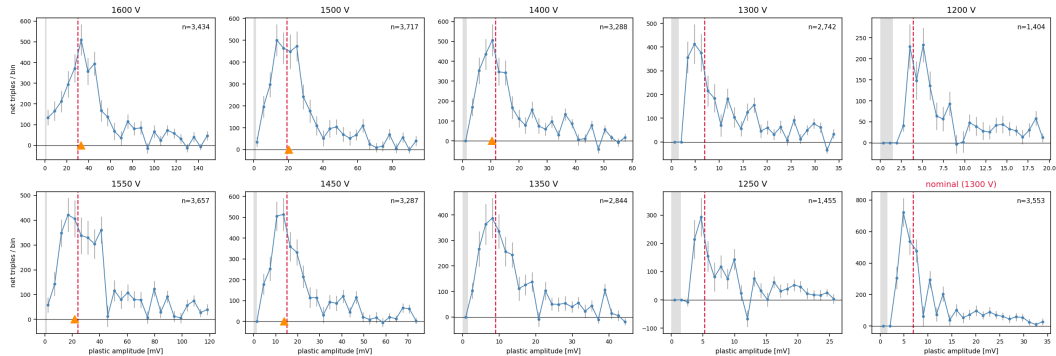
With LIQ readout (want today): wall $\times$ plastic $\times$ LS-1
(track must pass THROUGH the plastic — true MIP selection for the plastic)



Stack: MM $\rightarrow$ wall $\rightarrow$ plastic $\rightarrow$ LS. Wall $\times$ plastic tags wall MIPs only; adding a LIQ hit *behind* the plastic selects through-goers $\Rightarrow$ plastic spectrum in triples = plastic MIP candidates. Double sideband subtraction in both dt's.

Result 4 — the plastic MIP, found (PSSB2, one HV step per panel)

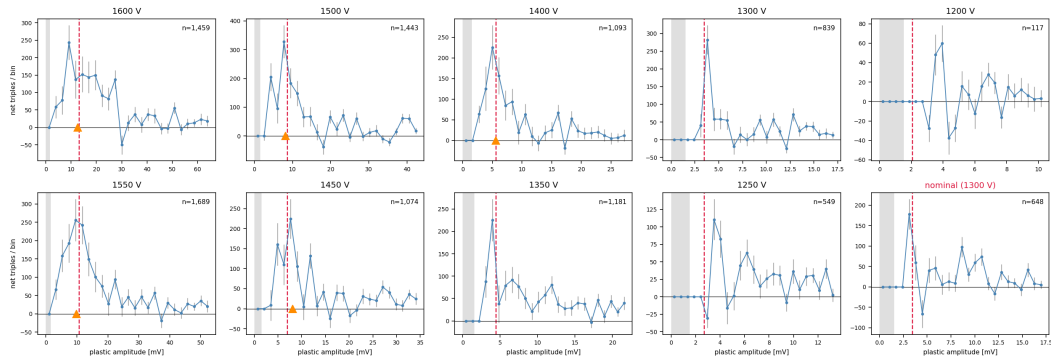
run224489 PSSB2: triple-tagged plastic spectrum per HV step (linear axis, x-range 0.5x expected MIP; dashed = power-law-expected MIP, gray = acquisition threshold)



Linear amplitude axis, coarse bins, propagated Poisson errors. A single significant peak tracks the power-law expectation (dashed) panel after panel while the acquisition threshold (gray) stays put — a threshold artifact cannot do this. Orange: per-step fitted MPV. All 8 PMTs in [figures/19_triples/steps_linear/](#).

Weakest PMT (PSSD2): clear at high V, rides threshold at nominal

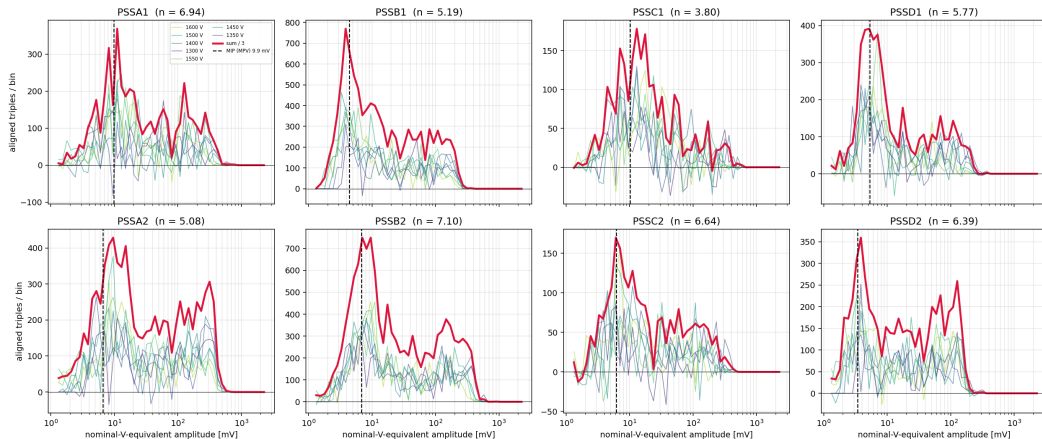
run224489 PSSD2: triple-tagged plastic spectrum per HV step (linear axis, x-range 0..5x expected MIP; dashed = power-law-expected MIP, gray = acquisition threshold)



Peak clear of threshold only for $V \geq 1400$ V $\Rightarrow$ only those steps enter the fit; at nominal HV the DR MIP rides the threshold — this arm cannot trigger on plastic MIPs until equalized.

Cross-check: gain-aligned stack

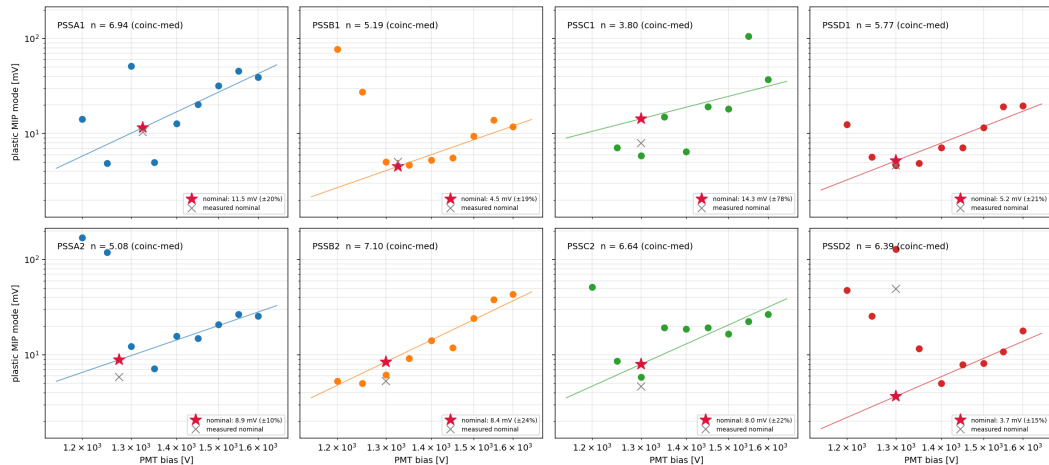
run224489: gain-aligned triple spectra — steps pile up at a common peak (dashed: adopted MIP) while the threshold edge moves



Each step's spectrum scaled to its nominal-V equivalent ($\times (V_{\text{nom}}/V)^n$ = translation in log amp): the steps pile up at a common peak and the sum (red) peaks at the adopted MPV (dashed) on every PMT. (Log-amp representation — peak positions here sit above the linear MPV.)

Plastic MIP vs HV $\rightarrow$ calibration at nominal

run224489: plastic MIP mode vs HV (power-law fits, star = extrapolation to nominal)



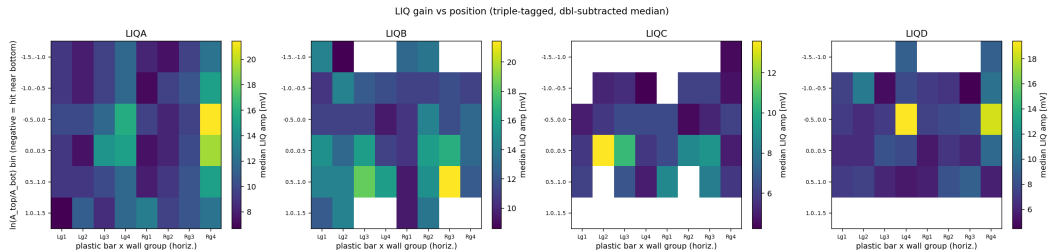
First plastic-MIP calibration table (linear-space MPV)

PMT	MPV [mV]	spread	mV/MeV
AL	9.9	11%	1.96
AR	6.6	31%*	1.31
BL	4.4	7%	0.87
BR	6.9	12%	1.37
CL	10.3	26%	2.04
CR	6.2	22%	1.23
DL	5.4	8%	1.07
DR	3.5	6%	0.69

* AR's steps split into two groups — flagged. Energy scale: 5.05 MeV in 2.5 cm PVT (normal incidence).

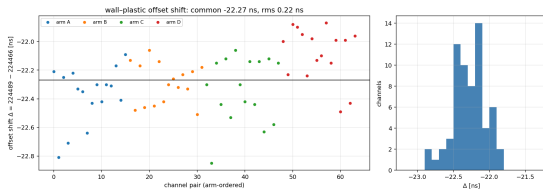
- ▶ Error-weighted MPV fits on the linear per-step spectra ($V \geq 1400$ V), transported to nominal V with each PMT's coincident-median n .
- ▶ First-pass log-binned modes were 15–35% high (mode of $dN/d \log A \neq$ MPV of dN/dA) — corrected; spreads improved to 6–12% for five PMTs.
- ▶ MIP-based inter-PMT ratios agree with the equalization table — independent cross-check.
- ▶ DR (3.5 mV) and BL (4.4 mV) MPVs sit at/below the 4.9 mV trigger threshold $\Rightarrow$ equalize before trusting plastic triggers on those arms.

Result 5 — LIQ gain-vs-position gradient is **real**

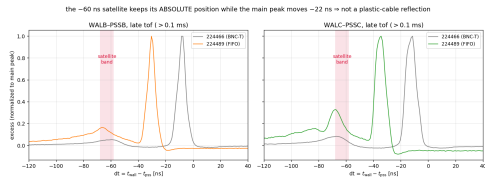


Triple-tagged track position: horizontal = wall group $\times$ bar, vertical = wall top/bottom asymmetry.
1.5–2 $\times$ gradient toward the PMT; axis flips with surveyed orientation: horizontal on A/D (PMT toward group 4), vertical on B/C (PMT up) — matches the as-built Geant geometry.

Timing: one clean common delay; the satellite misbehaves



All 64 wall-plastic offsets shifted by a **common** **-22.27 ns (rms 0.22 ns)**.



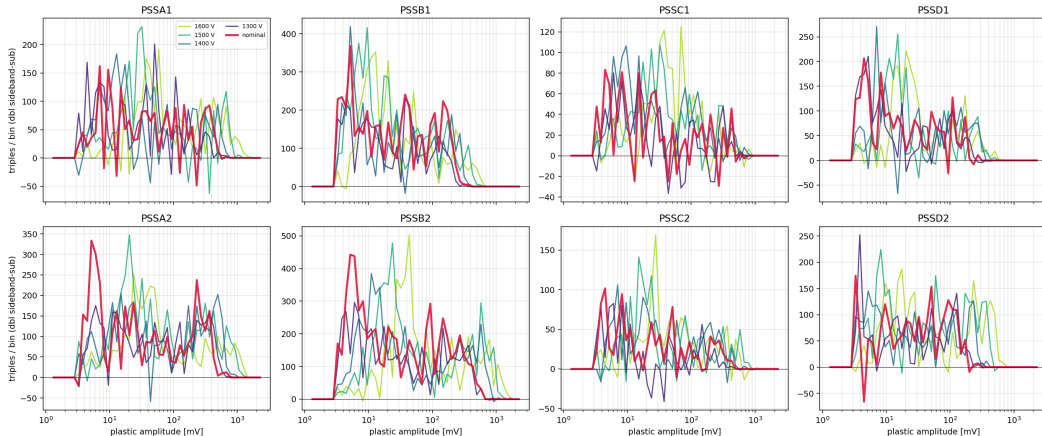
The -60 ns satellite kept its *absolute* position (main peak moved -22 ns) and grew to 19–34% of the peak $\Rightarrow$ not a plastic-cable reflection; injected at/after the FIFO. **Scope check recommended.** (Outside all analysis windows.)

Summary

1. **A/D cross-wiring fixed and confirmed** — the old A/D deficit is fully explained; wall MIP calibration now covers all four arms.
2. **First plastic MIP calibration**: 0.69–2.04 mV/MeV at nominal HV via WAL×PSS×LIQ triples.
3. **FIFO ratio measured**: $\times 1.40$ fleet (1.13–1.65 per PMT), not $\times 2$.
4. **New HV equalization** ready (DAQ repo); apply it — DR/BL cannot trigger on plastic MIPs at 4.9 mV until then.
5. **Liquids timed in**; LIQ position-gain gradient ($1.5\text{--}2\times$) confirmed and mapped; C/D LIQ gains look low — review LIQ HV.
6. **Scope-check the FIFO** for the -60 ns satellite.

Backup: per-step triple spectra (low stats per step)

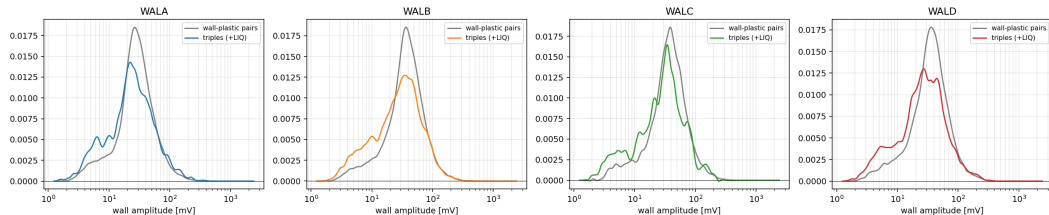
run224489: triple-tagged (WALxP55xLIQ) plastic spectra per HV step (rebinned x6)



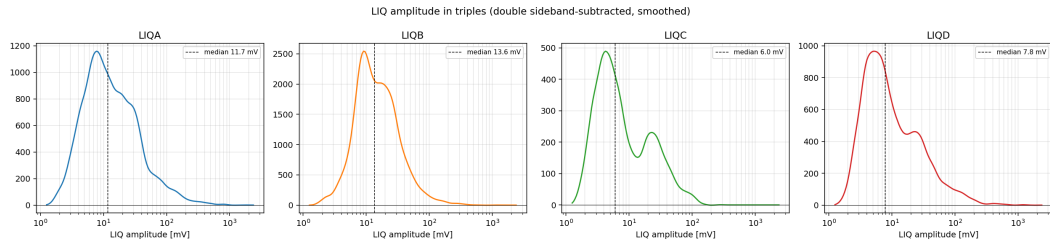
~2–6 k net triples per PMT per step after double sideband subtraction — individually noisy; the gain-aligned stack is the evidence plot, these show the raw marching with HV.

Backup: wall spectrum unchanged by the LIQ tag

wall spectrum: adding the LIQ tag does not move the wall MIP (normalized, smoothed)



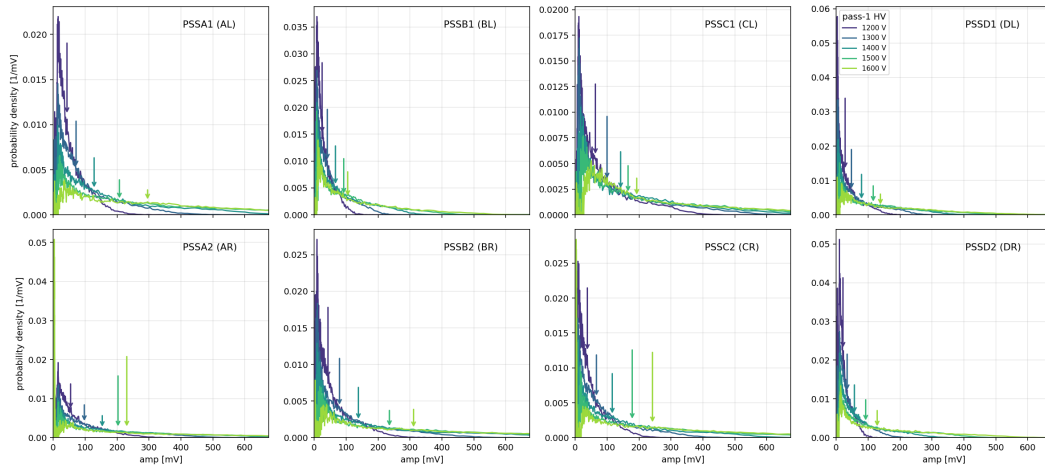
Backup: triple-tagged LIQ spectra (C/D low)



Inclusive LIQ medians: A 7.5, B 10.4, C 4.7, D 5.3 mV.

Backup: coincident spectra per step (linear mV)

run224489: wall-tagged coincident plastic spectra vs HV, linear amplitude axis (pass 1, gating OFF); arrows mark each spectrum's median = the equalization observable



Backup: trigger threshold scan at equalized gains

run224489: plastic threshold scan, equalized operating point (no signal/noise valley: efficiency-driven choice)

